

Confusion Matrix

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

- False Negative : also known as Type I error
- False Positive : also known as Type II error

Metrics

- Accuracy shows how many predictions were right out of all predictions:

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN}$$

- Precision focus on quality of positive predictions:

$$\text{Precision} = \frac{TP}{TP + FP}$$

- Recall measures how good the model is at predicting positives:

$$\text{Recall} = \frac{TP}{TP + FN}$$

- F1-Score combines precision and recall to balance their trade-offs.

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$